

DSA Course Content - RRGILucknow

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B.Tech Second Year

1: Introduction to DSA & Java Basics

- Introduction to Data Structure
- Introduction to Algorithm
- Types of Data Structures
- Linear vs Non-linear Data Structures
- Primitive vs Non-Primitive Data Structures
- Algorithm Design Steps
- Time Complexity
- Space Complexity
- Big O, Omega, Theta Notation
- Best, Average, Worst Case Analysis
- Debugging of Program

Java Basics: - Introduction to Java, JDK/JRE/JVM - Setting up IDE, writing & running first program - Variables & Data Types (int, double, char, boolean, etc.) - Operators - Taking input using Scanner - if/else, switch - for, while, do-while loops - Functions/Methods (declaration, parameters, return type) - Classes & Objects, Constructors - Basic OOP concepts (needed for later topics like Comparable, Comparator, Employee/Student objects)

2: Arrays

- Introduction to Arrays
- Memory Representation
- Single-Dimensional Arrays
- Multi-Dimensional Arrays
- Dynamic Arrays (ArrayList)
- Array Traversal
- Insertion
- Deletion
- Searching
- Updating
- Rotation
- Prefix Sum
- Two Pointer
- Sliding Window
- Largest Element
- Second Largest
- Missing Number
- Duplicate Elements
- Reverse Array
- Rotate Array
- Leaders in Array
- Move Zeroes
- Merge Arrays
- Linear Search
- Binary Search

- Kadane's Algorithm

3: Sorting Algorithms

- Bubble Sort
- Selection Sort
- Insertion Sort
- Merge Sort
- Quick Sort
- Sort Employee Objects
- Sort Student Marks
- Frequency Sort
- Merge Two Sorted Arrays

4: Strings

- String Basics
- Character Arrays
- StringBuilder
- StringBuffer
- String Methods
- Pattern Matching Basics
- Reverse String
- Palindrome
- Count Vowels
- Frequency Count
- Remove Duplicates
- Longest Word
- Anagram Check

5: Recursion & Backtracking Basics

- Introduction to Recursion
- Recursive Thinking
- Base Case
- Stack Memory
- Tail Recursion
- Backtracking Introduction
- Factorial
- Fibonacci
- Power Function
- Sum of Digits
- Reverse Number
- Generate Subsets (Basic)

6: Linked List

- Introduction
- Node Structure
- Singly Linked List
- Doubly Linked List
- Circular Linked List
- Traversal
- Insertion
- Deletion
- Searching
- Reverse Linked List
- Find Middle Node
- Detect Loop (Basic)
- Merge Lists
- Remove Duplicates

7: Stack & Queue

- Stack using Array
- Stack using Linked List
- Applications
- Queue using Array
- Queue using Linked List
- Circular Queue
- Deque
- Priority Queue
- Parenthesis Matching
- Undo Operation
- Expression Evaluation

8: Hashing

- Hash Function
- Collision Handling
- HashMap
- HashSet
- LinkedHashMap
- TreeMap
- Frequency Count
- First Non-Repeating Character
- Duplicate Detection
- Two Sum

9: Trees

- Introduction to Trees
- Binary Tree
- Binary Search Tree
- Tree Terminology
- Traversals
 - Inorder
 - Preorder
 - Postorder
 - Level Order
- Tree Height
- Count Nodes
- Search in BST
- Insert into BST
- Delete Node
- Find Middle Node
- Detect Loop (Basic)
- Merge Lists
- Remove Duplicates

10: Graph Basics

- Graph Terminology
- Directed & Undirected Graph
- Weighted Graph
- Graph Representation
- Adjacency Matrix
- Adjacency List
- Breadth First Search (BFS)
- Depth First Search (DFS)
- Connected Components
- Path Existence

11: Java Collections Framework for DSA

- ArrayList
- LinkedList
- Stack
- Queue
- PriorityQueue
- HashMap
- HashSet
- TreeMap
- TreeSet
- Comparable
- Comparator
- Sorting Objects
- Student Management
- Inventory System

12: Problem Solving & Coding Practice

- Coding Techniques
- Pattern-Based Problems
- Dry Run Techniques
- Debugging
- Input Optimization
- Arrays
- Strings
- Linked List
- Stack
- Queue
- Hashing
- Trees
- Graphs

1. Arrays

Problem	Difficulty	Link
Two Sum	Easy	https://leetcode.com/problems/two-sum/
Best Time to Buy and Sell Stock	Easy	https://leetcode.com/problems/best-time-to-buy-and-sell-stock/
Contains Duplicate	Easy	https://leetcode.com/problems/contains-duplicate/
Running Sum of 1D Array	Easy	https://leetcode.com/problems/running-sum-of-1d-array/
Merge Sorted Array	Easy	https://leetcode.com/problems/merge-sorted-array/
Remove Duplicates from Sorted Array	Easy	https://leetcode.com/problems/remove-duplicates-from-sorted-array/
Move Zeroes	Easy	https://leetcode.com/problems/move-zeroes/
Squares of a Sorted Array	Easy	https://leetcode.com/problems/squares-of-a-sorted-array/
Maximum Subarray (Kadane)	Medium	https://leetcode.com/problems/maximum-subarray/
Rotate Array	Medium	https://leetcode.com/problems/rotate-array/

2. Strings

Problem	Difficulty	Link
Valid Anagram	Easy	https://leetcode.com/problems/valid-anagram/
Valid Palindrome	Easy	https://leetcode.com/problems/valid-palindrome/
Reverse String	Easy	https://leetcode.com/problems/reverse-string/
Longest Common Prefix	Easy	https://leetcode.com/problems/longest-common-prefix/
Isomorphic Strings	Easy	https://leetcode.com/problems/isomorphic-strings/
Reverse Words in a String	Medium	https://leetcode.com/problems/reverse-words-in-a-string/

3. Binary Search

Problem	Difficulty	Link
Binary Search	Easy	https://leetcode.com/problems/binary-search/
Search Insert Position	Easy	https://leetcode.com/problems/search-insert-position/
Guess Number Higher or Lower	Easy	https://leetcode.com/problems/guess-number-higher-or-lower/
First Bad Version	Easy	https://leetcode.com/problems/first-bad-version/
Find Peak Element	Medium	https://leetcode.com/problems/find-peak-element/

4. Linked List

Problem	Difficulty	Link
Reverse Linked List	Easy	https://leetcode.com/problems/reverse-linked-list/
Middle of the Linked List	Easy	https://leetcode.com/problems/middle-of-the-linked-list/
Merge Two Sorted Lists	Easy	https://leetcode.com/problems/merge-two-sorted-lists/
Linked List Cycle	Easy	https://leetcode.com/problems/linked-list-cycle/
Remove Duplicates from Sorted List	Easy	https://leetcode.com/problems/remove-duplicates-from-sorted-list/
Remove Linked List Elements	Easy	https://leetcode.com/problems/remove-linked-list-elements/

5. Stack

Problem	Difficulty	Link
Valid	Easy	https://leetcode.com/problems/valid-

Parentheses		parentheses/
Implement Stack using Queues	Easy	https://leetcode.com/problems/implement-stack-using-queues/
Min Stack	Medium	https://leetcode.com/problems/min-stack/
Baseball Game	Easy	https://leetcode.com/problems/baseball-game/

6. Queue

Problem	Difficulty	Link
Implement Queue using Stacks	Easy	https://leetcode.com/problems/implement-queue-using-stacks/
Design Circular Queue	Medium	https://leetcode.com/problems/design-circular-queue/

7. HashMap & HashSet

Problem	Difficulty	Link
Two Sum	Easy	https://leetcode.com/problems/two-sum/
Majority Element	Easy	https://leetcode.com/problems/majority-element/
Ransom Note	Easy	https://leetcode.com/problems/ransom-note/
Happy Number	Easy	https://leetcode.com/problems/happy-number/
Intersection of Two Arrays	Easy	https://leetcode.com/problems/intersection-of-two-arrays/

8. Trees

Problem	Difficulty	Link
Maximum Depth of Binary Tree	Easy	https://leetcode.com/problems/maximum-depth-of-binary-tree/
Same Tree	Easy	https://leetcode.com/problems/same-tree/
Invert Binary Tree	Easy	https://leetcode.com/problems/invert-binary-tree/
Binary Tree Inorder Traversal	Easy	https://leetcode.com/problems/binary-tree-inorder-traversal/
Symmetric Tree	Easy	https://leetcode.com/problems/symmetric-tree/
Path Sum	Easy	https://leetcode.com/problems/path-sum/

9. Graphs

Problem	Difficulty	Link
Find if Path Exists in Graph	Easy	https://leetcode.com/problems/find-if-path-exists-in-graph/

Flood Fill	Easy	https://leetcode.com/problems/flood-fill/
Number of Islands	Medium	https://leetcode.com/problems/number-of-islands/
Clone Graph	Medium	https://leetcode.com/problems/clone-graph/

10. Recursion

Problem	Difficulty	Link
Fibonacci Number	Easy	https://leetcode.com/problems/fibonacci-number/
Climbing Stairs	Easy	https://leetcode.com/problems/climbing-stairs/
Pow(x, n)	Medium	https://leetcode.com/problems/powx-n/
Generate Parentheses	Medium	https://leetcode.com/problems/generate-parentheses/

Bonus Interview Problems

Problem	Difficulty	Link
Merge Intervals	Medium	https://leetcode.com/problems/merge-intervals/
Product of Array Except Self	Medium	https://leetcode.com/problems/product-of-array-except-self/
Top K Frequent Elements	Medium	https://leetcode.com/problems/top-k-frequent-elements/
Kth Largest Element in an Array	Medium	https://leetcode.com/problems/kth-largest-element-in-an-array/
Group Anagrams	Medium	https://leetcode.com/problems/group-anagrams/
Container With Most Water	Medium	https://leetcode.com/problems/container-with-most-water/
3Sum	Medium	https://leetcode.com/problems/3sum/
Longest Substring Without Repeating Characters	Medium	https://leetcode.com/problems/longest-substring-without-repeating-characters/